Reinforcement Learning for Inventory Management using Deep Q Networks

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***Abstract*—Inventory management is a critical decision-making problem in supply chain systems where retailers must determine optimal replenishment policies under uncertain demand and operational constraints. Traditional analytical models typically assume that demand distributions and system parameters are known in advance, which limits their applicability in dynamic real-world environments.**

In this work, the inventory control problem is formulated as a Markov Decision Process (MDP) and solved using reinforcement learning techniques. A Deep Q Network (DQN) is used as the baseline implementation to learn ordering policies through interaction with a simulated inventory environment. The system models stochastic demand, ordering costs, holding costs, and stockout penalties.

Experimental results demonstrate that the reinforcement learning agent is capable of learning adaptive ordering policies that balance inventory availability with operational costs. The framework can potentially be extended and compared with other reinforcement learning algorithms such as Double DQN and Actor–Critic methods.

I. Introduction

Inventory management plays an important role in supply chain operations. Retailers must determine how much inventory to order in order to meet uncertain customer demand while minimizing operational costs. Excess inventory leads to increased holding costs and storage constraints, whereas insufficient inventory results in lost sales and dissatisfied customers.

Traditional inventory models such as Economic Order Quantity (EOQ) and base-stock policies assume that demand distributions are known and stationary. However, real-world demand patterns are often stochastic and may change over time, making these assumptions unrealistic.

Reinforcement Learning (RL) provides a framework for sequential decision making in uncertain environments. Instead of relying on known demand models, RL agents learn optimal policies through interaction with the environment. In this work, we model the inventory control problem as a Markov Decision Process and apply a Deep Q Network (DQN) to learn optimal ordering decisions.

II. Literature Survey

Inventory control has been extensively studied in operations research. Classical approaches often rely on analytical policies derived under simplifying assumptions such as stationary demand and known system parameters. One of the most well-known results in inventory theory is the optimality of base-stock policies for certain stochastic inventory systems.

However, real-world inventory systems often involve complexities such as stochastic demand, lead times, and lost sales, which make analytical solutions difficult to compute. As a result, researchers have explored approximate dynamic programming and heuristic approaches to address these challenges.

Recent advances in machine learning have enabled the application of reinforcement learning techniques to inventory control problems. Reinforcement learning formulates the problem as a sequential decision-making process in which an agent learns an optimal policy through trial-and-error interaction with the environment.

Several studies have demonstrated that deep reinforcement learning algorithms can effectively learn inventory policies in stochastic environments. Dehaybe et al. [1] show that reinforcement learning can learn near-optimal ordering policies for stochastic lot-sizing problems. Similarly, Amiri et al. [2] study inventory systems with non-stationary demand and propose adaptive algorithms for learning optimal policies over time.

Other work has also explored reinforcement learning for inventory optimization in complex supply chains, including multi-echelon systems and lost-sales models [3]. These studies highlight the potential of reinforcement learning to handle complex decision spaces that are difficult to solve using classical optimization methods.

Despite these advances, reinforcement learning approaches still face challenges related to training stability and exploration. Nevertheless, RL remains a promising approach for learning adaptive policies in inventory management problems.

III. Problem Formulation

We consider a single-item inventory system where decisions are made at discrete time steps. At the beginning of each day, the retailer decides how many units of inventory to order. Orders placed on day t arrive on day $t+1$, introducing a one-day lead time.

The inventory control problem is modeled as a Markov Decision Process defined by the tuple

$$(S, A, P, R, \gamma)$$

where S represents the state space, A represents the action space, P represents the state transition function, R represents the reward function, and γ is the discount factor.

A. State

The state of the system at time t is defined as

$$s_t = (I_t, P_t, \hat{D}_t)$$

where I_t represents the current inventory level, P_t represents pipeline inventory arriving the next day, and $\hat{D}_t$ represents a noisy forecast of demand.

B. Action

The action corresponds to the number of units ordered by the retailer. For example, actions may be restricted to multiples of a fixed pack size:

$$a_t \in \{0, 12, 24, \ldots, A_{max}\}$$

In practice, the action space can be adjusted depending on system constraints.

C. Demand Model

Customer demand is modeled as a stochastic process. For illustration, demand may follow a normal distribution:

$$D_t \sim N(50, 10^2)$$

However, the demand distribution can be modified depending on the experimental setup used in the simulation.

D. Reward Function

The reward represents the daily profit:

$$r_t = pS_t - ca_t - hI_{t+1} - \phi L_t$$

where p represents selling price, c represents ordering cost, h represents holding cost, and ϕ represents the stockout penalty.

The objective is to maximize the expected cumulative discounted reward

$$\max E\left[\sum_{t=0}^{\infty} \gamma^t r_t\right]$$

IV. Baseline Implementation

As a baseline reinforcement learning method, this work uses the Deep Q Network (DQN) algorithm.

DQN is a value-based reinforcement learning algorithm that approximates the optimal action-value function $Q(s, a)$ using a neural network. The network takes the current state as input and outputs Q-values for all possible actions.

The Q-learning update rule is

$$Q(s, a) \leftarrow Q(s, a) + \alpha\left[r + \gamma \max_{a'} Q(s', a') - Q(s, a)\right]$$

To stabilize training, two key techniques are used:

- Experience Replay
- Target Networks

Experience replay stores past transitions and samples them randomly during training to reduce correlation between updates. Target networks provide stable Q-value targets by periodically updating a separate network.

V. Proposed Idea

The proposed idea is to apply reinforcement learning to learn adaptive inventory replenishment policies directly from interaction with the environment.

The primary implementation in this study uses a Deep Q Network as the baseline algorithm. The reinforcement learning agent observes the current inventory state, selects an order quantity, and updates its policy based on the reward received from the environment.

In addition to the baseline DQN implementation, the framework can potentially be extended to include other reinforcement learning algorithms. Two promising approaches include:

- **Double Deep Q Networks (Double DQN)** which reduce overestimation bias in Q-learning.
- **Actor–Critic methods** which use separate networks to learn the policy and value function.

Future work may compare these methods to evaluate their effectiveness for inventory management problems.

VI. Conclusion

This work demonstrates how reinforcement learning can be applied to inventory management problems. By modeling the system as a Markov Decision Process and applying Deep Q Networks, the agent learns adaptive ordering policies directly from interaction with the environment.

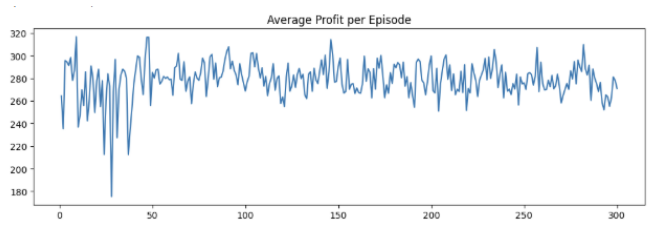
The results(Fig. 1) show that reinforcement learning can effectively balance ordering costs, holding costs, and stockout penalties in stochastic demand environments. While the current implementation focuses on the DQN algorithm, future work could explore extensions such as Double DQN and Actor–Critic methods for improved performance.

References

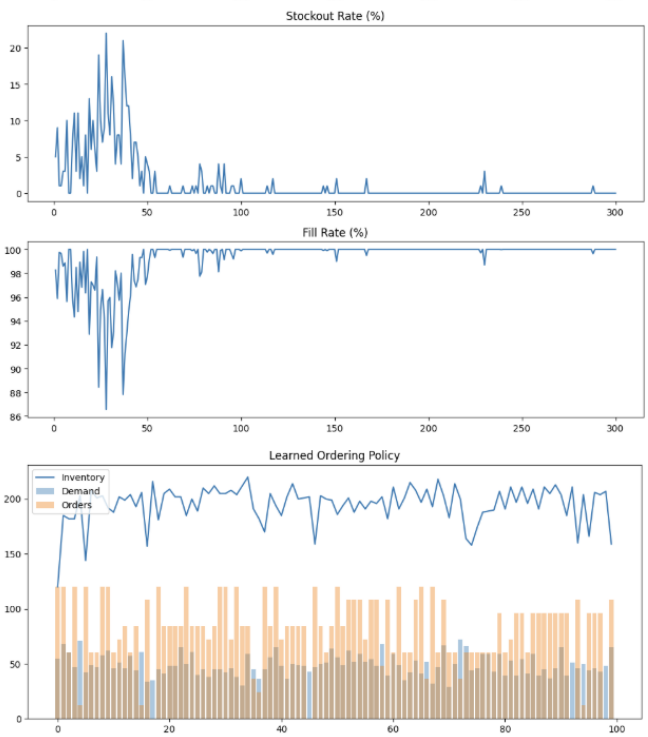
[1] H. Dehaybe, D. Catanzaro, and P. Chevalier, "Deep reinforcement learning for inventory optimization with non-stationary uncertain demand," *Preprint submitted to Elsevier*, 2024.

[2] N. H. Amiri, S. R. Sinclair, and M. Udenio, "Non-stationary inventory control with lead times," *arXiv preprint arXiv:2602.05799*, 2026.

[3] J. Gijsbrechts, R. N. Boute, J. A. V. Mieghem, and D. J. Zhang, "Can deep reinforcement learning improve inventory management? performance on lost sales, dual sourcing and multi-echelon problems," *SSRN Electronic Journal*, 2019.



(a) DQN Training Performance



(b) Inventory Level Over Time

Fig. 1: Performance of the DQN-based inventory control agent